Interpretability & XAI

From Black Boxes to Transparent Decisions

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Outline

1. Why XAI?
2. Families of Explainability Methods
3. LIME
4. SHAP
5. Grad-CAM
6. Probing
7. Summary

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What is Explainable AI (XAI)?

Interpretability

Intrinsic property

The model structure is directly human-readable — no external tool needed.

Examples: linear regression, decision trees

Explainability

Post-hoc

A black-box model is described after training via an external method.

Examples: SHAP, LIME, Grad-CAM

Interpretability Refresher

Linear Regression

Each coefficient β_i directly quantifies the effect of feature x_i on the output. The prediction is a weighted sum — every contribution is explicit and additive.

$$\hat{y} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

Decision Trees

A prediction is produced by following a sequence of human-readable if/else rules from root to leaf. The full reasoning path is visible and can be stated in plain language.

e.g. Age $>$ *30 & Income* $\leq$ *50k* $\Rightarrow$ *Class A*

Why Does XAI Matter?

Applications

- **Healthcare**
- **Finance**
- **Autonomous systems**
- Trust, accountability & safety
- Regulatory compliance (e.g. GDPR, AI Act)

Technical value

- **Debugging**: find data and/or modelling errors
- **Scientific discovery**: gain better insights

Why High Accuracy Is Not Enough? Shortcut Learning

Definition

A model minimises training loss by relying on features **irrelevant to the true task** — spurious correlations instead of genuine patterns.

Real-world example

Skin-cancer classifier: learned to detect **ruler marks** in dermoscopy images. Rulers appeared more often next to malignant lesions in training data.

Takeaway

High accuracy $\neq$ correct reasoning.
XAI catches shortcut effects *before* deployment.

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White-box vs. Black-box Models

White-box models

- Structure & parameters fully **accessible**
- Examples: linear regression, logistic regression, decision trees
- Trade-off: typically *less powerful* on complex tasks
- Term is sometimes used interchangeably with "interpretable" models

Black-box models

- Only input–output behavior exposed
- Deep neural networks, proprietary APIs
- State-of-the-art performance
- Require **post-hoc** explainability methods

Reality is a spectrum

Shallow NNs, regularised ensembles, attention visualization — occupy an **intermediate** space. The choice involves a practical trade-off between predictive power and understanding.

Local vs. Global Explanations

Local explanations

Why did the model produce **this** *output for* **this** *input?*

- Valid only in the neighborhood of one data point
- LIME: linear surrogate around a single input
- Grad-CAM: per-image heatmap

Global explanations

What patterns has the model learned **in general***?*

- Summarise behavior over the full input distribution
- Trustworthy regions of model's decision space
- Failure mode areas to avoid
- Aggregated feature importance rankings

Main Categories of XAI Methods

Perturbation-based
LIME
Occlusion maps

Attribution-based
SHAP
DeepLIFT

Gradient-based
Integrated Gradients
Grad-CAM

Concept-based
Probing classifiers
TCAV

All four families are **complementary** — use tools from multiple categories.

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LIME — Core Idea

Core insight

Even if a model is globally complex, its behavior can be approximated **locally** around any input by a simple surrogate model.

Three steps

1. **Perturb** the input x to generate $\{z_1, \ldots, z_n\}$
2. **Query** the black box f and weight by proximity $\pi_x(z_i)$
3. **Fit** a simple interpretable surrogate g^* to approximate f locally

LIME — Strengths & Limitations

Strengths

- **Model-agnostic**: works with any classifier
- Applicable across **images, text, tabular** data
- Query-only: works even on proprietary APIs

Limitations

- Explanation valid **only locally**
- Quality depends on perturbation strategy
- **Unstable**: running twice may yield different attributions

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SHAP — Core Idea

Game-theory foundation

Features are **players**; the model's prediction is the **payoff**. The Shapley value of feature i is its average marginal contribution across all possible feature subsets:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} \big[f(S \cup \{i\}) - f(S) \big]$$

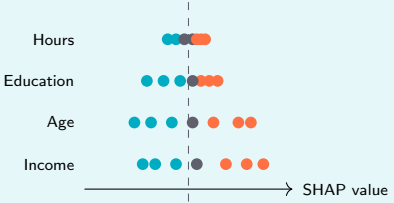
Practical variants

KernelSHAP LIME-like sampling; provably recovers Shapley values

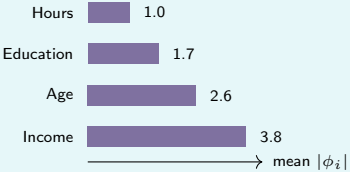
TreeSHAP Exact, polynomial-time for tree-based models

SHAP — Visualisations

Beeswarm plot



Bar chart (mean $|\phi_i|$)



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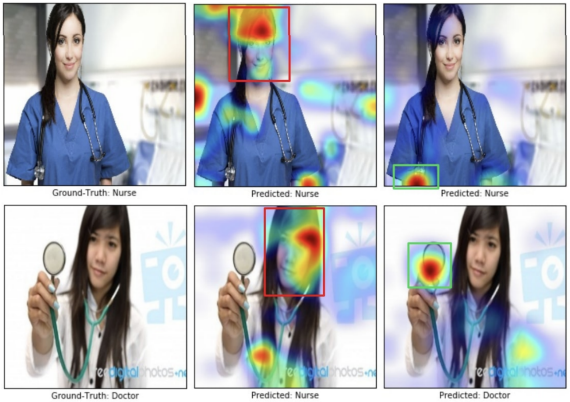
Grad-CAM — Core Idea

Core insight

The last convolutional layer retains **spatial information**; gradients of the class score y^c w.r.t. those feature maps encode **which locations matter most**.



Grad-CAM — Strengths & Limitations

Strengths

- **No retraining** required
- **Class-discriminative**: different class $\Rightarrow$ different heatmap
- Single forward + backward pass — **computationally cheap**

Limitations

- Inherits spatial resolution of last conv. layer
- CNN-specific (not model-agnostic)

Practical use

Verify the model attends to the **right** image regions (stethoscope) vs. spurious ones (face).

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Probing Classifiers — Core Idea

What is probing?

Train a simple classifier (the **probe**) on top of a frozen intermediate representation to test whether a specific concept is **linearly decodable** from that layer.

Three steps

1. **Extract** activations $h^{(\ell)}$ from layer ℓ of the frozen model for a labelled dataset
2. **Train** a lightweight classifier to predict the concept label from $h^{(\ell)}$
3. **Evaluate** probe accuracy — high accuracy $\Rightarrow$ the concept is encoded in that layer

Probing Classifiers — Strengths & Limitations

Strengths

- Reveals **what** information is encoded layer-by-layer
- Non-invasive: the original model is never modified
- Applicable to any architecture that exposes intermediate activations

Limitations

- High probe accuracy shows a concept is *present*, not that the model *uses* it for its predictions
- Results depend on the choice of probe complexity and layer
- Requires a labelled concept dataset, which can be costly to obtain

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Summary & Takeaways

The big picture

1. **High accuracy is not enough**.
2. The white-box / black-box distinction determines which tools are applicable.
3. Local & global explanations are **complementary**.
4. Four major families: gradient, perturbation, attribution, concept.

Three workhorse algorithms

LIME
model-agnostic, local

SHAP
principled, local+global

Grad-CAM
CNN, spatial, gradient

Key message

Use XAI not as an afterthought, but as a core part of the ML pipeline

Thank you for your attention!

Questions?